

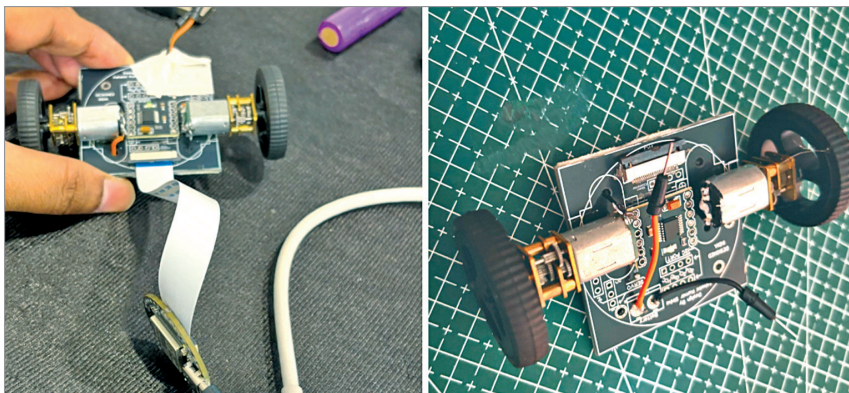


Fig. 6: Testing the robot motor controller and chassis

modules such as lidar units, radar sensors, cameras, robotic arms, and other custom peripherals.

This arrangement results in a compact, modular, and highly expandable robotic platform. Fig. 3 shows the PCB layout of the motor-driver chassis.

SMD assembly

Once the PCB design is completed (Fig. 3), the board is sent for fabrication. After receiving the manufactured PCBs, assembly begins with mounting the N20 motors in their designated locations and soldering their electrical connections.

Next, the DRV8833 motor-driver IC and other SMD components, including capacitors and resistors, are assembled using standard surface-mount soldering techniques. The FPC connector, servo connectors, expansion headers, and remaining interface connectors are then soldered in place to complete the hardware assembly.

After assembly, the PCB forms a compact and fully integrated robot chassis that combines motor control, power distribution, and expansion interfaces on a single board. Fig. 4

shows the fabricated PCB and completed SMT assembly.

Construction and testing

After fabrication, the N20 geared motors were mounted in the designated slots, followed by soldering the DRV8833 motor driver, FPC connector, expansion headers, and other SMD components. The wheels were then attached to the motor shafts, completing the compact robot chassis assembly.

The motor driver is compatible with a wide range of microcontrollers and can be tested directly using the IndusBoard Coin. For testing, select the ESP32S2 Dev Module in the Arduino IDE, choose the appropriate COM port, and upload the program. The IndusBoard Coin is connected to the robot motor-driver PCB through an FPC cable, allowing the motors to operate according to the uploaded code. A source-code snippet is shown in Fig. 5.

Fig. 6 illustrates the construction and testing of the integrated motor-driver robot chassis. The image on the left shows the chassis connected to the controller through the FPC

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cable during operation, while the image on the right shows the fully assembled prototype with motors and wheels fitted.

To verify performance, an ESP32-based controller was interfaced through the FPC connector and programmed to execute forward, reverse, left, and right movements. All functions operated successfully, confirming reliable communication through the FPC interface and proper operation of the DRV8833 motor driver. The tests also demonstrated the advantages of the integrated chassis design, including reduced wiring complexity and simplified assembly.

For future expansion, a dedicated microcontroller board with a compatible FPC connector can be designed according to the required I/O assignments. Alternatively, the IndusBoard Coin can be connected directly through the FPC cable to control motors, displays, sensors, and servo motors. If the FPC interface is not required, the motor driver can also be connected directly to a controller via the onboard header pins using the provided solder-jumper configuration. **EFY**

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